

UTTI, Finland

WMO: 029660

Lat: 60.891N

Lon: 26.941E

Elev: 103

StdP: 100.09

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-24.0	-20.8	-26.5	0.3	-23.8	-23.0	0.5	-20.6	9.3	-3.4	8.5	-3.6	2.3	340	0.630

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.9	27.4	18.2	25.7	17.4	23.9	16.6	19.8	25.0	18.7	23.5	17.7	22.1	4.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	13.1	21.6	17.0	12.3	20.5	15.9	11.4	19.5	57.2	24.9	53.5	23.5	50.1	22.1	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.4	6.6	-26.2	29.4	4.1	2.1	-29.2	30.9	-31.6	32.1	-33.9	33.3	-36.9	34.8		
(5)			-26.1	21.3	4.3	1.3	-29.2	22.2	-31.6	23.0	-34.0	23.7	-37.1	24.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.9	-6.8	-6.8	-2.9	3.9	10.1	14.8	17.8	16.1	10.9	4.8	0.0	-3.8		
	DBStd	9.77	6.68	6.64	4.51	3.83	4.34	3.52	3.28	3.00	3.35	3.91	4.51	6.06		
	HDD10.0	2561	522	472	399	188	53	3	0	1	31	164	301	428		
	HDD18.3	4963	781	705	657	434	257	117	50	83	224	419	550	687		
	CDD10.0	697	0	0	0	4	55	147	243	188	57	3	0	0		
	CDD18.3	59	0	0	0	0	2	11	34	13	0	0	0	0		
	CDH23.3	488	0	0	0	0	23	97	275	92	0	0	0	0		
	CDH26.7	85	0	0	0	0	3	12	59	11	0	0	0	0		
(14) Wind	WSAvg	3.3	3.3	3.3	3.4	3.2	3.4	3.3	3.1	3.0	3.1	3.3	3.4	3.6		
(15) Precipitation	PrecAvg	649	51	41	39	30	36	60	69	71	58	69	65	55		
	PrecMax	857	97	97	78	57	84	107	134	150	130	145	142	87		
	PrecMin	490	11	4	5	9	10	14	7	6	18	14	8	15		
	PrecStd	90	22	23	19	13	23	25	34	36	36	33	31	20		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.9	4.7	9.8	19.5	26.1	28.2	30.5	28.0	21.6	14.1	9.3	7.5		
		MCWB	3.6	2.4	5.0	11.3	15.4	17.7	19.5	18.3	15.7	11.8	8.7	6.9		
	2%	DB	2.8	2.7	6.1	16.0	22.9	26.0	28.3	25.8	19.3	12.5	8.1	5.5		
		MCWB	2.4	1.9	2.8	9.2	14.2	16.6	19.1	17.6	14.6	11.2	7.7	5.1		
	5%	DB	1.9	1.7	4.0	13.2	20.4	24.0	26.3	24.0	17.8	11.3	7.0	4.2		
		MCWB	1.7	1.3	1.6	7.7	12.6	16.2	18.2	16.9	13.8	10.3	6.7	3.8		
	10%	DB	0.9	0.8	2.5	10.7	18.1	21.9	24.3	22.0	16.1	10.2	5.9	3.0		
		MCWB	0.7	0.5	1.0	6.1	11.4	14.9	17.3	16.2	13.4	9.4	5.6	2.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.5	3.0	5.4	12.8	17.5	19.9	21.5	20.5	16.8	12.8	8.9	6.8		
		MCDB	3.6	3.6	8.7	17.5	23.1	25.5	26.8	25.5	19.5	13.5	9.3	7.1		
	2%	WB	2.3	2.0	3.3	10.3	15.3	18.5	20.3	18.8	15.7	11.7	7.7	5.2		
		MCDB	2.4	2.5	5.3	15.0	20.0	23.7	25.8	22.9	18.0	12.3	8.0	5.4		
	5%	WB	1.5	1.2	2.2	8.3	13.7	17.1	19.4	17.8	14.8	10.7	6.7	3.8		
		MCDB	1.7	1.6	3.4	12.3	19.1	21.9	24.6	22.0	16.9	11.3	6.9	4.1		
	10%	WB	0.6	0.4	1.2	6.6	12.1	15.7	18.3	16.9	13.8	9.6	5.6	2.7		
		MCDB	0.7	0.7	2.1	10.0	17.3	20.6	23.0	20.7	15.7	10.1	5.8	2.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.1	5.7	7.2	8.7	10.1	9.0	8.9	8.6	7.4	5.2	4.0	4.5		
		MCDBR	3.8	4.5	8.7	13.1	13.4	12.3	11.6	11.5	9.4	6.0	4.4	3.9		
	5% WB	MCWBR	3.9	4.1	6.1	7.5	6.6	5.7	4.8	4.8	5.4	4.6	4.4	3.8		
		MCWBR	3.8	4.1	6.9	11.9	11.7	10.6	10.3	9.6	8.0	5.5	4.3	3.5		
(40) Clear-Sky Solar Irradiance	taub	taub	0.262	0.302	0.326	0.364	0.361	0.357	0.380	0.368	0.344	0.325	0.287	0.373		
		taud	2.106	2.167	2.236	2.335	2.418	2.458	2.414	2.437	2.496	2.403	2.187	2.612		
	Edn at Noon	Edn at Noon	463	667	789	829	863	872	839	819	779	654	430	316		
		Edh at Noon	40	71	93	103	102	100	102	92	74	57	38	26		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.24	0.85	2.33	3.81	5.17	5.41	5.38	4.11	2.41	1.00	0.27	0.13		
	RadStd	RadStd	0.05	0.18	0.29	0.41	0.57	0.34	0.51	0.45	0.28	0.16	0.05	0.02		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (52 neighbors)	+0.50	N/A	N/A	N/A	N/A	N/A	-161	-171	N/A	N/A

Nomenclature: See separate page